

Expository Report on Density and Bias

Ethan Sosin, Annalaura Pegoraro and Jingyi Le

July-August 2025

1 Introduction

The motivation for this expository report comes from the paper *On the number of places of convergence for Newton's method over number fields* by Faber and Voloch, which analyzes the outcome of Newton's method in the non-archimedean setting. The authors conjecture that the set of primes p for which Newton's iteration fails to p -adically converge modulo p has full density, and also provide two numerical examples, which we aim to investigate in more detail.

We begin by outlining the setting of the paper. Let f be a non-constant polynomial with coefficients in a number field K , in our case it will be $\mathbb{Q}$, and let $x_0 \in K$. We define the Newton map

$$N(t) = N_f(t) = t - \frac{f(t)}{f'(t)},$$

and for each $n \geq 0$ define (x_n) to be the sequence given by $x_{n+1} = N(x_n)$. If $\alpha \in \overline{K}$ is a root of f , we say it is an *exceptional* root if the Newton sequence (x_n) converges v -adically to α for at most finitely many places v of K .

The main result by Faber and Voloch is the following.

Theorem 1.1. *Let $f \in K[t]$ be a polynomial of degree $d \geq 2$ and let $x_0 \in K$. Assume (x_n) is not eventually periodic. Then:*

- (1) *There exists a finite set of places S of K , depending only on the polynomial f , s.t. if $v \notin S$, then either (x_n) converges v -adically to a simple root of f or else (x_n) does not converge in $\mathbb{P}^1(K_v)$. In particular, any multiple root of f is exceptional.*
- (2) *Suppose α is a simple root of f . Then α is exceptional if and only if N_f is totally ramified at α .*
- (3) *The sequence (x_n) diverges in $\mathbb{P}^1(K_{\mathfrak{p}})$ for infinitely many places v of K .*

Building on this result, the authors propose a conjecture concerning the density of places where Newton's iteration converges to a root of f .

Conjecture 1.4. *Assume the same hypotheses of Theorem 1.1. Let $C(K, f, x_0)$ be the set of places v of K for which (x_n) converges v -adically to a root of f . Then the natural density of the set $C(K, f, x_0)$ is zero.*

To support this conjecture, the authors provide two examples of cubic polynomials and highlight additional features of independent interest. The first example provided by Faber and Voloch is given by the polynomial $f(t) = t^3 - 1$, which has no exceptional roots. Defining the density

$$\delta(x_0, X) = \frac{\{\#p \leq X : (x_n) \text{ converges to a root of } f \text{ in } \mathbb{Q}_p\}}{\pi(X)},$$

they analyse its behaviour for $x_0 = 2, 3, 4, 5$ and N up to 200,000. Expecting it to tend to zero as predicted by the density conjecture, the authors observe that the values of $\delta(x_0, X)$ are decreasing with X , but somewhat slowly. In Section 2 we extend the authors' computations and study some additional examples, considering both Newton and Non-Newton maps.

The second example is given by the polynomial $g(t) = t^3 - t$, for which the only exceptional root is $\alpha = 0$. In order to determine if one of the roots ± 1 is a limit of the sequence (x_n) more often than the other, the authors define the quantities

$$\delta_+(x_0, X) = \frac{\{\#p \leq X : x_n \rightarrow +1 \text{ in } \mathbb{Q}_p\}}{\pi(X)},$$

$$\delta_-(x_0, X) = \frac{\{\#p \leq X : x_n \rightarrow -1 \text{ in } \mathbb{Q}_p\}}{\pi(X)},$$

and compute them for $x_0 = 2, 3, 4, 5$ and X up to 100,000. The data seem to suggest a bias toward the root $+1$, but it is somewhat ambiguous. In section 3 we intend to explore this phenomenon further.

Finally, we explain how the code used to reproduce and extend these results works. In particular, in the proof of Theorem 1.1 (2), it is proved that if α is a simple root and $\mathfrak{p}$ is a prime ideal outside of a certain finite set of prime ideals such that $\mathfrak{p} \mid (x_n - \alpha)$ for some n , then (x_n) converges to α in the $\mathfrak{p}$ -adic topology. More concretely, after discarding finitely many primes p , the sequence (x_n) converges in $\mathbb{P}^1(\mathbb{Q}_p)$ to a simple root α iff $x_n \equiv \alpha \pmod{p}$ for some n , or equivalently $f(x_n) \equiv 0 \pmod{p}$ for some n . We also recall that any multiple root of f is exceptional. Finally, as stated in Corollary 1.2, Faber and Voloch prove that all roots of a quadratic polynomial are exceptional.

These results help us outline a pathway to implement the numerical code. Suppose now $f \in \mathbb{Q}[t]$ is a polynomial of degree $d \geq 3$ with no repeated roots and $x_0 \in \mathbb{P}^1(\mathbb{Q})$. After discarding finitely many primes, we can assume that N has good reduction modulo p . If we consider the reduced map $\tilde{N} : \mathbb{P}^1(\mathbb{F}_p) \rightarrow \mathbb{P}^1(\mathbb{F}_p)$, the orbit $(x_n \pmod{p})$ is well defined and eventually becomes periodic. As noted above, for any prime p outside of a certain finite set, the sequence (x_n) converges in $\mathbb{Q}_p$ to a simple root α iff $x_n \equiv \alpha \pmod{p}$ for some n . If that is the case, this means that the orbit $(x_n \pmod{p})$ eventually encounters a fixed point of $\tilde{N}$. Note also that the converse must be true. Assuming good reduction and tracing through the proof of Corollary 3.2, we can see that if we encounter a fixed point γ of the reduction $\tilde{N}$, then $x_n = \gamma \equiv \alpha \pmod{p}$ for some α root of f and for some n , hence the sequence converges. Accordingly, our code is based on finding the fixed points of the reduced map $\tilde{N}$.

2 Density of Primes yielding p -adic Convergence

2.1 Newton Maps

In this section we aim to study the behaviour of the density of primes for which Newton's method converges for $f(t) = t^3 - 1$ and other cubic polynomials. The objective is to gather evidence that may either support or challenge the conjecture, and to examine if similar patterns emerge across different cases.

The Newton map associated to the polynomial $f(t) = t^3 - 1$ is

$$N_f(t) = \frac{2t^3 + 1}{3t^2}.$$

For $p = 2, 3$ the map is not well defined, so we need to treat these cases separately. For $p = 2$ the sequence converges for any starting point x_0 , while for $p = 3$ we have convergence when $x_0 \equiv 1 \pmod{3}$. With this in mind, we can now generate Table 1 overleaf, which extends the data of Faber and Voloch tenfold.

Perhaps, a more effective way to understand the behaviour of δ is to plot its values. Let us consider the case $x_0 = 2$. By generating a log-log plot of the data in Figure 1 below (plotting $\log \delta(X)$ against $\log X$), the trend appears to be linear.

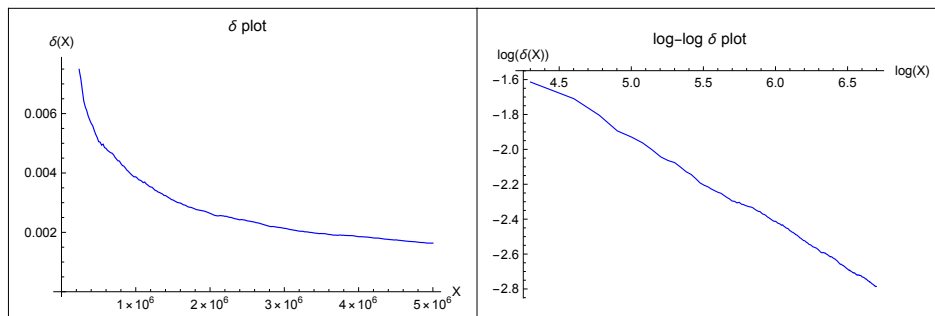


Figure 1: Plot and log-log plot of $\delta(X)$ for $x_0 = 2$ and X ranging up to 5,000,000 in increments of 20,000.

Therefore, we proceed with a linear regression on the log-log data, and compute the coefficient of determination R^2 . The result is a relation of the form $\log(\delta(X)) = m \log(X) + q$, where the estimated parameters are $m \approx -0.5019$ and $q \approx 0.5829$. The coefficient of determination is very close to 1 ($R^2 \approx 0.9982$), suggesting that the density follows a decay of the form

$$\delta(X) \sim 10^q X^m \text{ as } X \rightarrow \infty.$$

We may then ask if this behaviour holds for all starting points. Figures 2 and 3 overleaf shows the values of the coefficients q and m obtained for 200 initial points ranging from -100 to $+100$, with the exclusion of $x_0 = 0$, for which the Newton map is not defined, and $x_0 = 1$, which is a root of f .

X	Starting point x_0			
	2	3	4	5
20K	2.431	2.476	2.962	2.962
40K	1.951	1.975	2.284	2.308
60K	1.568	1.634	1.800	1.816
80K	1.276	1.365	1.544	1.544
100K	1.178	1.209	1.376	1.345
120K	1.088	1.115	1.292	1.239
140K	0.9915	1.022	1.184	1.145
160K	0.9058	0.9467	1.062	1.069
180K	0.8628	0.9301	0.9852	1.016
200K	0.8396	0.9064	0.9119	0.9564
300K	0.6424	0.7347	0.7001	0.7616
400K	0.5670	0.6409	0.6084	0.6497
500K	0.5056	0.5633	0.5369	0.5706
600K	0.4827	0.5214	0.5031	0.5234
700K	0.4616	0.4793	0.4669	0.4934
800K	0.4316	0.4472	0.4316	0.4550
900K	0.4055	0.4237	0.4125	0.4307
1000K	0.3873	0.4000	0.3898	0.4038
1100K	0.3675	0.3792	0.3652	0.3815
1200K	0.3529	0.3637	0.3508	0.3669
1300K	0.3359	0.3469	0.3339	0.3449
1400K	0.3239	0.3361	0.3193	0.3323
1500K	0.3092	0.3224	0.3066	0.3180
1600K	0.2989	0.3096	0.2956	0.3071
1700K	0.2864	0.2997	0.2880	0.2981
1800K	0.2784	0.2895	0.2828	0.2873
1900K	0.2725	0.2844	0.2774	0.2809
2000K	0.2639	0.2746	0.2699	0.2739

Table 1: Table of values $100 \cdot \delta(x_0, X)$ for different starting points x_0 and primes $p \leq X$. We write 20K for 20,000. The first section of the table reproduces Faber-Voloch results; the second provides some extended data.

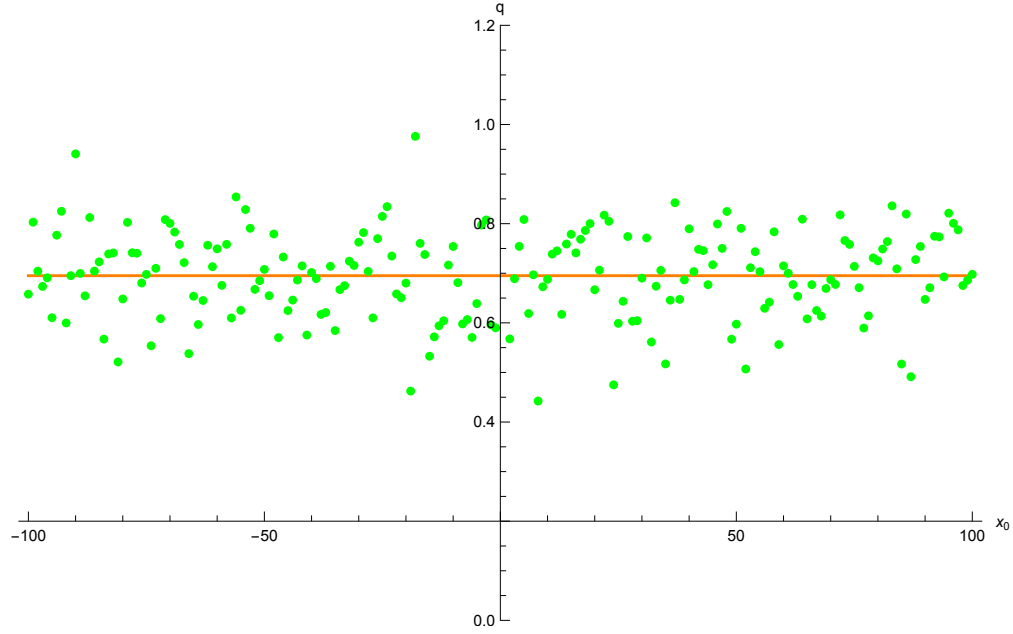


Figure 2: Plot of (x_0, q) , started at x_0 (horizontal) with q (vertical), obtained from X ranging up to 4,000,000 in increments of 20,000. The red line indicates the mean value, approximately 0.695105. We also note the standard deviation is approximately 0.0885095, indicating a small amount of variation around the mean.

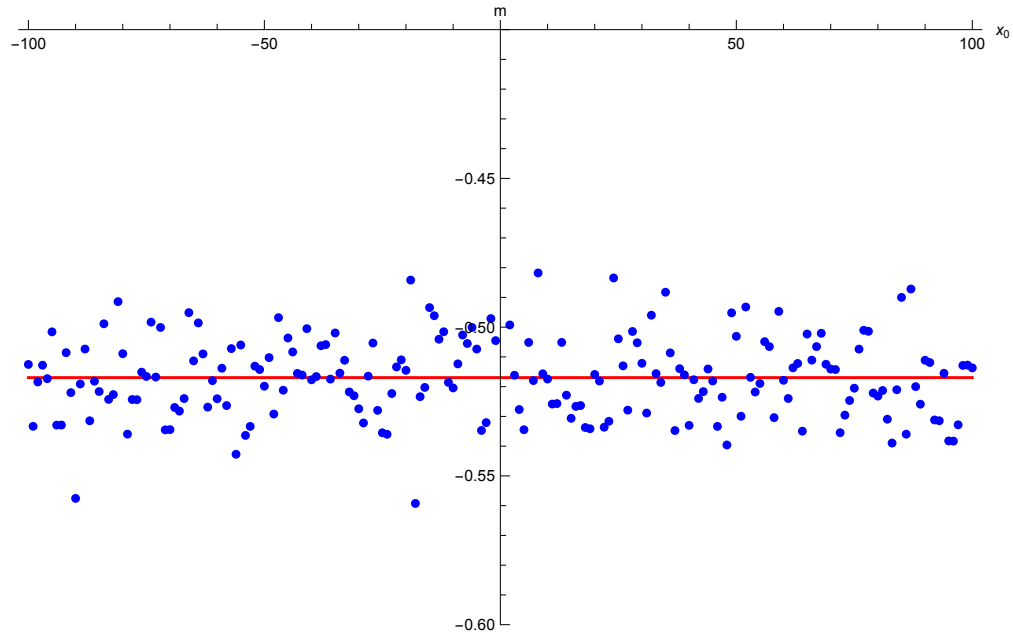


Figure 3: Plot of (x_0, m) , started at x_0 (horizontal) with m (vertical), obtained from X ranging up to 4,000,000 in increments of 20,000. The red line indicates the mean value, approximately -0.516955 . We also note the standard deviation is approximately 0.013476, indicating very little variation around the mean.

It can be readily seen that the values of m are close to $-1/2$. For each of these regressions, the coefficient of determination R^2 is greater than 0.9949, indicating a strong linear fit in the log-log data. This suggests that the observed relation

$$\delta(X) \sim \frac{10^q}{\sqrt{X}} \text{ as } X \rightarrow \infty$$

is not restricted to a specific starting point, but may be a more general phenomenon.

Of course, so far we have only considered the Newton map of one polynomial, and we would like to generalise our results to cubic polynomials of the form $f(t) = t^3 + at + b$, for $a, b \in \mathbb{Z}$, for which,

$$N_f(t) = \frac{2t^3 - b}{3t^2 + a}.$$

Let α be a root of f . Following the proof of the main theorem in Faber-Voloch,

$$N_f(t) - \alpha = (t - \alpha)^2 \left(\frac{2t + \alpha}{(t^2 + \alpha t + a + \alpha^2) + (t - \alpha)(2t + \alpha)} \right),$$

so we exclude prime ideals $\mathfrak{p} \mid 3\alpha^2 + a$. One can verify that

$$N_{\mathbb{Q}(\alpha)}(3\alpha^2 + a) = 4a^3 + 27b^2 = -\Delta(f),$$

therefore, we can apply our reduction mod p technique for all rational primes $p \nmid \Delta(f)$. In particular, we exclude pairs (a, b) for which $\Delta(f) = 0$.

For each choice of (a, b) with $-5 \leq a, b \leq 5$, we will generate all of the primes $p \leq 2,000,000$ (excluding finitely many as above) for which we have convergence of N_f , fixing $x_0 = 1$.

		Value of b										
		-5	-4	-3	-2	-1	0	1	2	3	4	5
Value of a	-5	-0.505	-0.475	-0.471	C	-0.557	D	-0.506	-0.454	-0.456	C	D
	-4	-0.480	-0.462	-0.549	-0.515	-0.498	C	-0.498	-0.528	C	D	-0.529
	-3	D	D	D	D	D	D	D	D	D	D	D
	-2	-0.510	-0.495	-0.460	-0.474	-0.487	-0.469	C	D	-0.460	C	-0.404
	-1	-0.480	-0.487	-0.454	-0.449	-0.503	C	-0.491	-0.446	-0.481	-0.487	-0.471
	0	-0.538	-0.507	-0.459	-0.533	C	C	-0.500	D	-0.536	-0.516	-0.538
	1	-0.474	-0.410	-0.501	C	-0.489	-0.525	-0.512	-0.425	-0.440	-0.480	-0.488
	2	-0.468	-0.486	C	-0.475	-0.538	-0.499	-0.539	-0.475	-0.508	-0.509	-0.516
	3	-0.514	C	-0.508	-0.467	-0.453	-0.478	-0.464	-0.467	-0.480	-0.453	-0.465
	4	C	-0.469	-0.495	-0.496	-0.501	-0.524	-0.503	-0.511	-0.538	-0.476	-0.480
	5	-0.417	-0.508	-0.439	-0.488	-0.558	-0.486	-0.497	-0.516	-0.449	-0.445	-0.485

Table 2: Estimates of m for the Newton map of $f(x) = x^3 + ax + b$ for $x_0 = 1$ and different values of a, b .

Note there are exceptional cases in the table - we have placed an 'C' in the table where we have p -adic convergence for all primes p , and a 'D' where we always have divergence; this can be seen by considering the map over the rationals.

It is unclear what to conclude from these results. The estimates for m are not too far from $-1/2$, but there are certainly outliers. We can be relatively confident of our hypothesis that $\delta(X) \sim CX^m$, since the R^2 values for each regression are all at least 0.975.

2.2 Non-Newton Maps

It is of interest to determine how much of the evidence given in Section 2.1 is specific to rational maps of Newton type. We would like to construct a family of rational maps that exhibit similar properties to those studied above; in particular, they should be cubic rational maps with some superattracting integer fixed points. We define the family,

$$f_c(t) = t - \frac{(t^2 - 1)(t^2 - c^2)}{t^3 - (2c^2 - 1)t}, \quad c \in \mathbb{Z}, \quad c \geq 2,$$

which has superattracting fixed points at ± 1 and repelling fixed points at $\pm c$. It is easy to check that this is not the Newton map of a polynomial, as the derivative of the numerator of $t - f_c(t)$ is not equal to its denominator. We also define a second non-Newton family,

$$g_c(t) = t - \frac{(t^2 - 1)(t^2 - c^2)}{t^3 + (c^2 - 2)t}, \quad c \in \mathbb{Z}, \quad c \geq 2,$$

which has superattracting fixed points at $\pm c$ and repelling fixed points at ± 1 .

For both of these families, we would like to understand for which primes we can employ the same convergence test we laid out in section 1 (that is, reducing the rational map modulo p and determining if the orbit of a point is eventually fixed). Fortunately, for superattractors, this technique works for all but finitely many primes. To see this for f_c , consider the expression,

$$\begin{aligned} f_c(t) \mp 1 &= (t \mp 1) \left(1 - \frac{(t \pm 1)(t^2 - c^2)}{t^3 - (2c^2 - 1)t} \right) \\ &= (t \mp 1) \left(\frac{\mp t^2 - (c^2 - 1)t \pm c^2}{t^3 - (2c^2 - 1)t} \right) \\ &= \mp(t \mp 1)^2 \left(\frac{t \pm c^2}{t^3 - (2c^2 - 1)t} \right) \\ &= \mp(t \mp 1)^2 \left(\frac{(t \mp 1) \pm (c^2 + 1)}{(t \mp 1)^3 \pm 3(t \mp 1)^2 - 2(c^2 - 2)(t \mp 1) \mp 2(c^2 - 1)} \right). \end{aligned}$$

So, if $p \nmid 2(c^2 - 1)$, then $p^n \mid t \mp 1 \Rightarrow p^{2n} \mid f_c(t) \mp 1$. Hence the reduced map being eventually fixed at $\pm 1 \pmod{p}$ is sufficient to conclude that we have p -adic convergence of f_c to ± 1 for all primes p not dividing $2(c^2 - 1)$.

Via a very similar calculation, one can show that the reduced map of g_c eventually reaching $\pm c \pmod{p}$ is sufficient to conclude that we have p -adic convergence of g_c to $\pm c$ for all primes p not dividing $2c(c^2 - 1)$.

Altogether, this means that we can run the same procedure as before on these non-Newton maps. Once again, we are interested in how the natural density $\delta(X)$ of primes $p \leq X$ with p -adic convergence to ± 1 decays. We shall again conjecture that $\delta(X) \sim CX^m$ as $X \rightarrow \infty$ for some constants m, C .

We will first test this hypothesis for $c = 2$ and $x_0 = 3$. The primes to exclude are those dividing $2(c^2 - 1) = 6$. So, we test all primes $5 \leq p \leq 4,000,000$ for p -adic convergence of f_c , and plot $\log \delta$ against $\log X$.

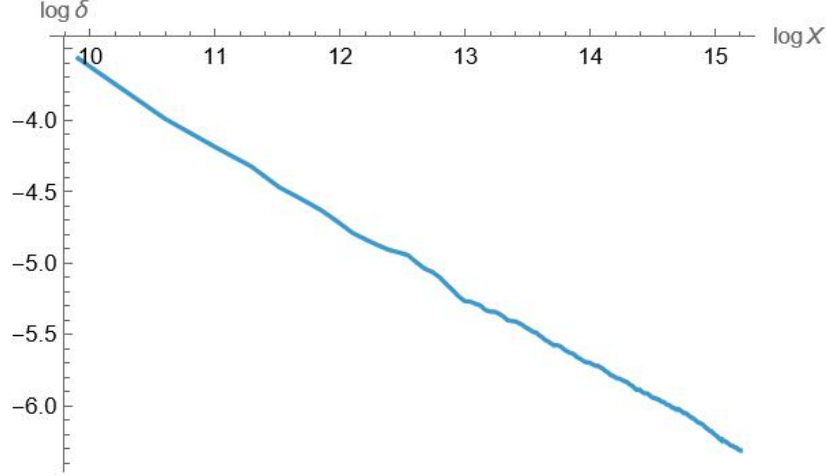


Figure 4: A plot of $\log(\delta(X))$ for $X \leq 4,000,000$ for the map f_2 with $x_0 = 3$.

The trend is visibly linear. Running a linear regression on this data, we get estimates $m \approx -0.496$ and $C \approx 3.485$. We also compute that the R^2 value for this regression is 0.9985. The next natural question to ask is whether our estimate for m depended on our choices of c and x_0 . We now run the same test for $2 \leq c, x_0 \leq 10$ for primes $p \leq 2,000,000$.

		Value of c								
		2	3	4	5	6	7	8	9	10
Initial Point x_0	2	N	-0.502	-0.481	-0.506	-0.482	-0.449	-0.517	-0.511	-0.523
	3	-0.499	N	-0.482	-0.487	-0.465	-0.466	-0.536	-0.502	-0.524
	4	N	-0.492	N	-0.500	-0.502	-0.470	-0.522	-0.492	-0.536
	5	-0.441	-0.484	-0.472	N	-0.506	-0.456	-0.465	-0.488	-0.524
	6	-0.506	-0.489	-0.452	-0.500	N	-0.454	-0.528	-0.509	-0.547
	7	-0.463	-0.521	-0.464	N	-0.499	N	-0.527	-0.502	-0.511
	8	-0.503	-0.531	-0.488	-0.510	-0.470	-0.481	N	-0.523	-0.527
	9	-0.478	N	-0.464	-0.518	-0.465	-0.438	-0.514	N	-0.517
	10	-0.478	-0.519	-0.477	-0.518	-0.515	-0.494	-0.511	-0.521	N

Table 3: Estimates of m for f_c for different initial points x_0 and values of c .

Note we have placed an ‘N’ in the table where x_0 is pre-fixed under f_c in the rationals, so p -adic convergence is of course independent of the choice of p . We have omitted the R^2 values from this table for clarity, but we note that, in all the cases without an ‘N’, the R^2 value is at least 0.993.

The estimates for $m(c, x_0)$ are all in the vicinity of $-1/2$. That said, the data is not as conclusive as it could be, with most notable outliers $m(7, 9) = -0.438$ and $m(10, 6) = -0.547$.

We now run the same tests on g_c , which we recall has superattracting fixed points at $\pm c$ and repelling fixed points at ± 1 . This time, we set $\delta(X)$ to be the natural density of primes $p \leq X$ for which we have p -adic convergence of g_c to $\pm c$. We again conjecture $\delta(X) \sim CX^m$ as $X \rightarrow \infty$ and test primes $p \leq 2,000,000$ to find an estimate for m .

		Value of c								
		2	3	4	5	6	7	8	9	10
Initial Point x_0	2	N	-0.505	-0.479	-0.482	-0.463	-0.513	-0.518	-0.472	-0.520
	3	-0.479	N	-0.479	-0.502	-0.477	-0.440	-0.518	-0.497	-0.507
	4	-0.492	-0.498	N	-0.502	-0.488	-0.496	-0.514	-0.525	-0.517
	5	-0.474	-0.477	-0.461	N	-0.500	-0.481	-0.487	-0.517	-0.519
	6	-0.487	-0.503	-0.482	-0.508	N	-0.439	-0.491	-0.523	-0.507
	7	-0.476	-0.468	-0.488	-0.490	-0.485	N	-0.519	-0.485	-0.536
	8	-0.462	-0.507	-0.480	-0.491	-0.447	-0.455	N	-0.483	-0.539
	9	-0.468	-0.491	-0.495	-0.504	-0.473	-0.455	-0.516	N	-0.528
	10	-0.454	-0.511	-0.464	-0.487	-0.513	-0.472	-0.498	-0.518	N

Table 4: Estimates of m for g_c for different initial points x_0 and values of c .

Once again, the estimates of m are all close to $-1/2$, albeit with some outliers, and we can be relatively confident of our log-log linear model, since the R^2 values this time are all at least 0.992.

This suggests that working towards proving that $\delta(X) \sim C/\sqrt{X}$ as $X \rightarrow \infty$ for superattractors of general cubic rational maps may be a fruitful direction of study.

3 Root Bias

In the section we intend to study the root bias phenomenon further.

First, we analyze the case $g(t) = t^3 - t$. Its associated Newton map is

$$N_g(t) = \frac{2t^3}{3t^2 - 1}.$$

Note that N_g has bad reduction for $p = 2$, hence we need to check this case by hand. In particular, the sequence converges to

$$\begin{cases} -1, & \text{for } x_0 = 3 \\ 0, & \text{for } x_0 = 2, 4 \\ 1 & \text{for } x_0 = 5 \end{cases}.$$

We can now reproduce and extend the results of Faber and Voloch, this time twenty-fold, on Table 5 overleaf.

X	Starting point x_0			
	2	3	4	5
20K	1.5473, 1.5031	1.5473, 1.1936	1.5031, 1.4147	1.5915, 1.1936
40K	1.0469, 0.9993	0.9755, 0.9041	0.9993, 0.9517	1.1420, 0.8327
60K	0.8915, 0.7925	0.8420, 0.7760	0.8255, 0.7760	0.9080, 0.7099
80K	0.7656, 0.6508	0.7273, 0.6763	0.7146, 0.7146	0.7784, 0.6252
100K	0.6568, 0.6151	0.6255, 0.6359	0.6568, 0.6151	0.6672, 0.5317
200K	0.4337, 0.4170	0.4782, 0.4615	0.4448, 0.4226	0.4615, 0.3726
300K	0.3577, 0.3154	0.4154, 0.3693	0.3423, 0.3500	0.3654, 0.3116
400K	0.2983, 0.2865	0.3692, 0.3308	0.2924, 0.3101	0.3249, 0.2806
500K	0.2769, 0.2624	0.3226, 0.2985	0.2672, 0.2841	0.2937, 0.2576
600K	0.2464, 0.2383	0.2872, 0.2668	0.2403, 0.2526	0.2668, 0.2403
700K	0.2246, 0.2193	0.2617, 0.2441	0.2264, 0.2264	0.2511, 0.2246
800K	0.2048, 0.2095	0.2439, 0.2236	0.2095, 0.2064	0.2299, 0.2095
900K	0.1978, 0.1950	0.2287, 0.2203	0.2020, 0.1964	0.2175, 0.2006
1000K	0.1822, 0.1847	0.2115, 0.2064	0.1949, 0.1847	0.2038, 0.1924
1100K	0.1727, 0.1797	0.2007, 0.1937	0.1843, 0.1773	0.1913, 0.1797
1200K	0.1657, 0.1732	0.1872, 0.1915	0.1775, 0.1668	0.1818, 0.1743
1300K	0.1590, 0.1650	0.1810, 0.1830	0.1690, 0.1650	0.1720, 0.1670
1400K	0.1540, 0.1568	0.1736, 0.1783	0.1606, 0.1615	0.1680, 0.1587
1500K	0.1454, 0.1507	0.1664, 0.1726	0.1524, 0.1594	0.1638, 0.1515
1600K	0.1412, 0.1453	0.1618, 0.1659	0.1494, 0.1552	0.1569, 0.1486
1700K	0.1358, 0.1420	0.1569, 0.1600	0.1428, 0.1561	0.1514, 0.1459
1800K	0.1347, 0.1392	0.1503, 0.1555	0.1399, 0.1525	0.1488, 0.1429
1900K	0.1303, 0.1366	0.1457, 0.1514	0.1352, 0.1486	0.1450, 0.1380
2000K	0.1282, 0.1316	0.1437, 0.1484	0.1316, 0.1457	0.1410, 0.1350

Table 5: Table of values $100 \cdot \delta_+, 100 \cdot \delta_-$ for different starting points x_0 and primes $p \leq X$. We write $20K$ for 20,000. The first section of the table presents Faber-Voloch results; the second provides some extended data.

Recall that 0 is the only exceptional root and the quantities defined for roots 1 and -1 are:

$$\delta_+(x_0, X) = \frac{\#\{p \leq X : x_n \rightarrow +1 \text{ in } \mathbb{Q}_p\}}{\pi(X)},$$

$$\delta_-(x_0, X) = \frac{\#\{p \leq X : x_n \rightarrow -1 \text{ in } \mathbb{Q}_p\}}{\pi(X)},$$

Faber and Voloch commented the following: *Most of the data suggests a bias towards the root $+1$ (most strongly for $x_0 = 5$), although we have no explanation at present for this behaviour.* From the extended data, it appears that what initially seemed to be a root bias towards $+1$ was in fact due to the values of X being too small. One way to investigate this further is to introduce the quantity

$$r_{x_0}(X) = \frac{\delta_+(x_0, X)}{\delta_-(x_0, X)},$$

the ratio of primes up to X converging to $+1$ over those converging to -1 . We can therefore produce a plot of this quantity for each starting point (Figure 5 below). The ratio values tend to remain close to 1, with the exception of the case $x_0 = 4$ which appears to stabilize towards 0.9, although the fluctuations appear somewhat erratic.

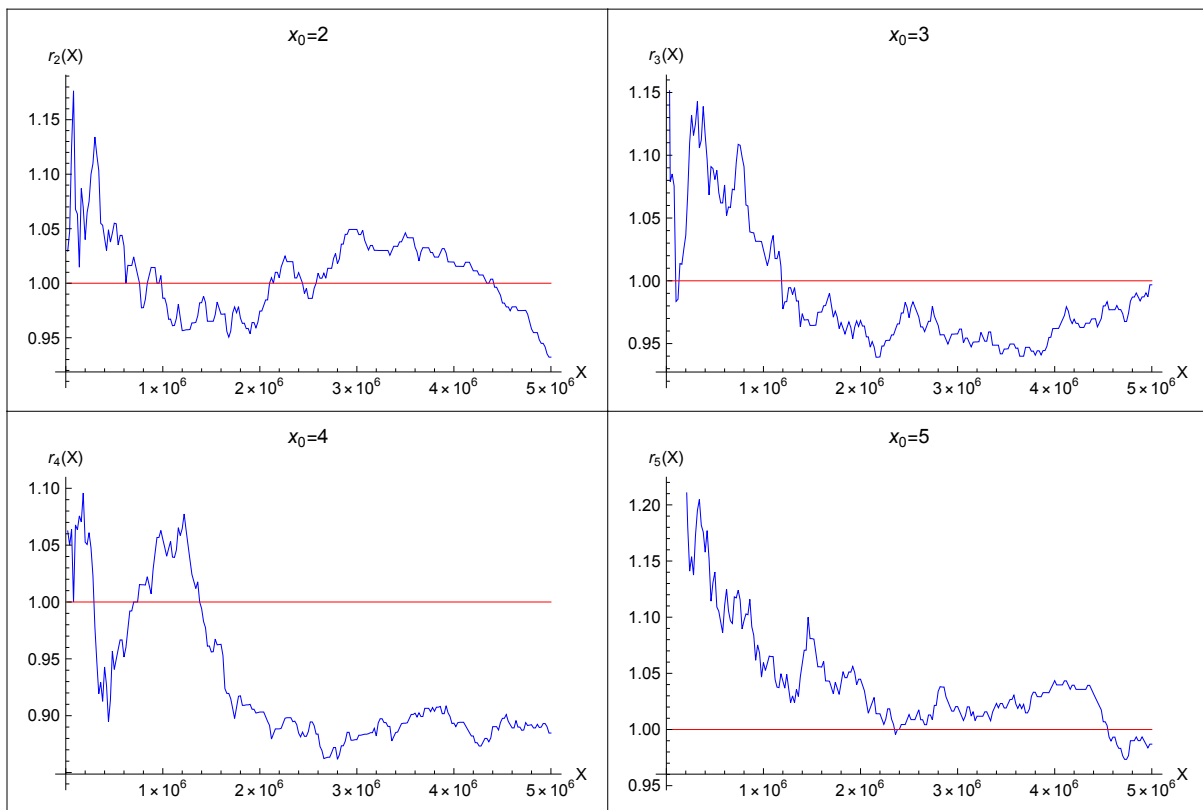


Figure 5: Plot of $r_{x_0}(X)$ for $x_0 = 2, 3, 4, 5$ and X ranging up to 5,000,000 in increments of 20,000.

We now run the same tests for $h(t) = t(t-1)(t-3)$, which has no exceptional roots. We need to exclude primes $p = 2, 3$. The results can be summarized in Figure 6 below.

The analysis was conducted on a set of 97 starting points. The cases -40 and 42 were included in Figure 6 because the densities for the different roots were more distinct in these instances. However, they still appear very similar. The evidence we have gathered from both polynomials points towards the conclusion that this is random noise, and thus that there is insufficient evidence to suggest that there is a root bias phenomenon.

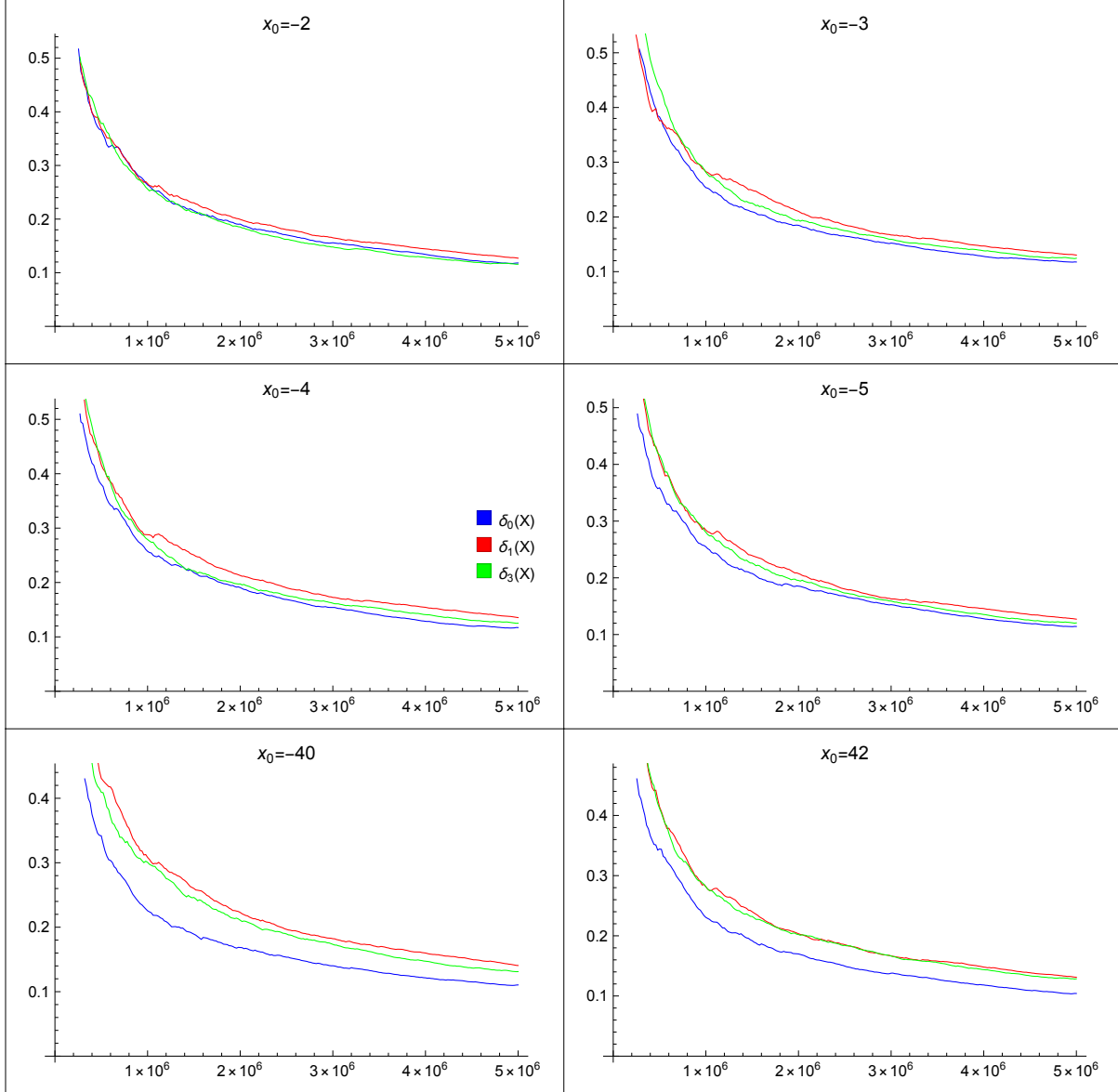


Figure 6: Plot of $\delta_\alpha(X) = \{\text{density of primes } \leq X \text{ converging to } \alpha\}$ for $x_0 = 2, 3, 4, 5, -40, 42$ and X ranging up to $5,000,000$ in increments of $20,000$.

4 Next Steps

Our study so far has focused on understanding the density and bias of primes for which Newton's method converges p -adically, particularly on cubic polynomials. Having explored both the Newton and non-Newton cases, the next natural stage is to broaden the scope to more general rational maps that extend or modify Newton's method. We now present these two promising directions.

First, the McMullen maps are defined via a modified Newton-type process: given a cubic polynomial $f(t) = t^3 + at + b$ and $h(t) = 3at^2 + 9bt - a^2$, one constructs the rational function

$$q(t) = \frac{f(t)}{h(t)}$$

and defines the associated map $T_f(t)$ as the Newton map of $q(x)$. That is,

$$T_f(x) = x - \frac{(x^3 + ax + b)(3ax^2 + 9bx - a^2)}{3ax^4 + 18bx^3 - 6a^2x^2 - 6abx - 9b^2 - a^3}. \quad (1)$$

This yields a new rational function of degree 4 if $\Delta(f) = -4a^3 - 27b^2 \neq 0$, otherwise it's a constant map. Compared to classical Newton maps of $f(t)$, it has different fixed points and its dynamics may display improved convergence behavior. Studying the mod p behavior of such maps - including period, fixed points, and root bias - would provide a broader context and possibly deep structural phenomena underlying Newton failure. Furthermore, we can attempt to recreate analogues of the proofs of Faber and Voloch with the McMullen map of a cubic polynomial.

Second, we consider quartic polynomials. As noted by McMullen's paper, Newton's method is generally convergent for quadratics, but not for cubics or higher degrees. In fact, he states that there is no purely iterative algorithm that is generally convergent for polynomials of degree four or more. This foundational limitation suggests that Newton-type algorithms are fundamentally obstructed for quartic and higher polynomials, so we have a strong motivation to examine the failure behavior in such cases. We may also ask whether a similar mod p convergence or density bias appears. Moreover, if they follow comparable patterns like $\delta(X) \sim C/\sqrt{X}$ as $X \rightarrow \infty$.

Together, these two directions - examining McMullen maps and extending to quartic polynomials - serve as natural and rigorous continuations of our current research. They also offer new views from which to approach the question of rational maps failing to find roots and may eventually enrich the theory of local and global dynamics in arithmetic settings.